

2) Use lump model

$$A = N \cdot 4\pi R^2 = \frac{m}{\rho_s \frac{4}{3}\pi R^3} \cdot 4\pi R^2 = \frac{3m}{\rho_s R} = 0.11 \text{ m}^2$$

$$m_s c_p \frac{dT_s}{dt} = -hA(T_s - T_\infty)$$

$$T_\infty = 20^\circ\text{C}$$

$$\frac{dT_s}{(T_s - T_\infty)} = \frac{-hA}{m_s c_p} dt$$

$$\ln\left(\frac{T_s - T_\infty}{T_{s,i} - T_\infty}\right) = t \frac{-hA}{m_s c_p}$$

$$t = \frac{m_s c_p}{hA} \ln\left(\frac{T_{s,i} - T_\infty}{T_s - T_\infty}\right)$$

$$t = \frac{400}{600(0.11)} \ln\left(\frac{40 - 20}{10 - 20}\right) = 70 \text{ s}$$

2) Sng

$$m_s c_p \frac{dT_s}{dt} = hA(T_s - T_w)$$

$$c_w$$

Sng

$$\frac{d}{dt}(c_s T_s + c_w T_w) = 0$$

$$\frac{dT_s}{dT_w} = -\frac{c_w}{c_s}$$

$$\int dT_s = \int -\frac{c_w}{c_s} dT_w$$



312

Home

Insert

Draw

View



BC Spelling

Switch Background

Immersive Reader

100%

Page Width



Password Protection



2) Say

$$m c_p \frac{dT_w}{dt} = h A (T_s - T_w)$$

Say

$$\frac{d}{dt} (c_s T_s + c_w T_w) = 0$$

No conversion ~ no energy

$$c_s T_s + c_w T_w = M \text{ constant}$$

$$T_w = \frac{M - c_s T_s}{c_w}$$

$$c_s T_s = -h A (T_s - T_w)$$

$$\frac{M - c_s T_s}{c_w}$$

$$= -h A \left(T_s - \frac{M}{c_w} + \frac{c_s T_s}{c_w} \right)$$

$$c_s T_s = -h A \left(T_s \left(1 + \frac{c_s}{c_w} \right) - \frac{M}{c_w} \right)$$

$$T_s + \frac{h A}{c_s} \left(1 + \frac{c_s}{c_w} \right) = \frac{h A M}{c_s c_w}$$

$$\frac{dT_s}{dT_w} = -\frac{c_w}{c_s}$$

$$\int dT_s = \int -\frac{c_w}{c_s} dT_w$$

$$T_{s1} - T_{s2} = -\frac{c_w}{c_s} (T_{w1} - T_{w2})$$



312

Home

Insert

Draw

View



BC Spelling

Switch Background

Immersive Reader

100%

Page Width



Password Protection



say



$$c_s (T_{si} - T_{sf}) = c_w (T_f - T_{wi})$$

$$T_f = \frac{c_s T_{si} + c_w T_{wi}}{c_s + c_w}$$

$$T_s(t) = T_f + (T_{si} - T_f) e^{-t/L}$$

find and plug to

get T_s

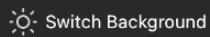
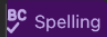
$$t = 27.5$$

1)

$$m_c c_{pc} = 51,000$$

$$\Delta T_c = 210 - 90$$

$$q = \Delta T_c m_c c_{pc} = 6 \times 10^5 \text{ Btu/h}$$



1)

$$m_c c_{p,c} = 5,000$$

$$\Delta T_c = 210 - 90$$

$$q = \Delta T_c m_c c_{p,c} = 6 \times 10^5 \text{ BTU/h}$$

$$q = m_h c_{p,h} \Delta T_h \quad \Delta T_h = 260 - X \quad \left. \begin{array}{l} \text{) } q_L \text{ equal} \\ \text{to } q_c \end{array} \right\} X = 200^\circ\text{F}$$

$$q = UA \frac{\Delta T_A - \Delta T_B}{\ln(\Delta T_A / \Delta T_B)} \quad \Delta T_A = T_{h,in} - T_{c,out}$$

$$A = 2\pi r L \cdot N_t \quad \Delta T_B = T_{h,out} - T_{c,in}$$

$$q = U \cdot 2\pi r L \cdot N_t \Delta T_{lm}$$

solve for L

$$L = \frac{2009}{U_o} \text{ ft?}$$

$$\text{if } U_o = 150$$

$$L = 13.4 \text{ ft}$$